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Unsteady RANS CFD simulations of ship manoeuvrability and course keeping control under various wave height conditions

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ABSTRACT

Ships encounter different wave heights in real sea states. Various wave heights can lead to substantial changes in the behaviour of a ship during manoeuvring, and hence a ship's manoeuvrability in waves should be accurately evaluated to ensure navigational safety at sea. The aim of this study is to estimate wave height effects on the manoeuvring behaviour of the benchmarking KRISO Container Ship (KCS) model by performing a fully nonlinear unsteady RANS simulation. Three types of manoeuvres were simulated with the free-running KCS model appended with an actuator disk and a moving semi-balanced horn rudder using a dynamic overset technique: self-propulsion, course keeping control, and turning circle manoeuvres. Manoeuvring analyses were carried out in the bow quartering waves of a range of wave heights for constant wave length, as coupled with the fifth-order Stokes wave model. In order to control the rudder deflection based on manoeuvre mechanisms, a feedback controller was embedded in the present CFD model. The results clearly revealed that wave heights have a strong effect on the manoeuvring behaviour of the KCS, including ship's speeds, seakeeping behaviour, and critical turning indices through comparative analyses under different wave height conditions. One important finding is the standard manoeuvring parameters in waves, which would help navigation officers in decision-making for manoeuvring actions in waves.

1. Introduction

With the increasing demand for marine transportation, shipping activities have accounted for more than 80% of global trade (Cepeda et al., 2019). The reliance on water-borne transportation has contributed to the number of maritime accidents claiming serious harm on life, property, and ocean environment. According to the Annual Overview of Marine Casualties and Incidents 2020 released by EMSA (2020), the navigation casualties associated with collision, contact and grounding incidents are responsible for more than 44% of all marine incidents. Undoubtedly, inadequate manoeuvring actions by navigation officers are recognised as the leading cause of such maritime accidents. Zhang and Li (2017) pointed out that rough sea conditions were closely associated with a great number of marine accidents. In addition, Ventikos et al. (2018) showed a close correlation between safe navigation and adverse weather conditions by carrying out the statistical analysis of navigational accidents related to the failure of manoeuvrability. Therefore, to improve navigational safety at sea, it is imperative to understand ships' manoeuvring behaviours in real sea states.

A ship's performance is highly dependent on sea conditions affected by waves. From a hydrodynamic point of view, sea waves affect ship motions and loads, as well as ship's manoeuvring performance. In other words, the manoeuvring behaviour can be affected by a combination of wave characteristics such as wave height, length, and directions when a ship sails in waves. Sea states can be practically described by the Beaufort scale proposed by the World Meteorological Organisation (from Beaufort 0 to Beaufort 12). It should be noted that Beaufort scale 12 defines waves of height greater than 14 m. In this regard, it is thought that the wave height could lead to substantial changes in the ship resistance and thus the loss of speed, which were highly expected to affect the manoeuvring characteristics of a ship in waves. It supports the argument of this paper that a proper understanding of the relationship between wave heights and ship manoeuvring behaviours in real sea conditions is necessary.

The manoeuvring performance of various ships in waves has been investigated by several approaches such as theoretical methods, experiments, and Computational Fluid Dynamics (CFD). As for theoretical methods, nonlinear mathematical models were proposed to estimate the

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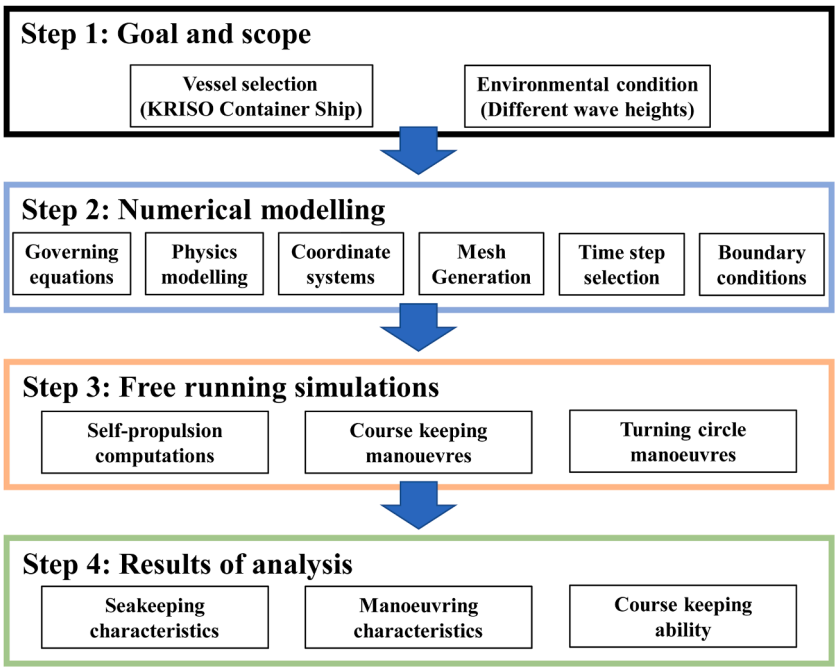


Fig. 1. Research methodology overview applied to the direct CFD manoeuvring simulations.

ship's manoeuvrability in waves by combining the potential flow theory. Fossen (2005) developed a unified state-space model for ship manoeuvring, station-keeping, and control in a seaway. However, this unified model has the underlying limitation that only some of the nonlinear wave effects are taken into account. Skejic and Faltinsen (2008) conducted a theoretical study of the combined seakeeping and manoeuvring of a ship in regular waves by developing a unified model based on the modular concept. Later on, several other studies have also applied similar theoretical methods to assess the ship's manoeuvrability in waves (Seo and Kim, 2011; Subramanian and Beck, 2015; Zhang and Zou, 2016; Paroka et al., 2017). However, those studies have revealed that analysis results obtained by the theoretical methods are highly likely to bring out some discrepancies with experimental data available. This is because that the reliability of the potential flow theory is limited due to the lack of physics associated with viscous effects and the free surface resolution. Therefore, those simplified models are not able to accurately capture the complex fluid-structure interactions between the hull, rudder, propeller, and waves, as discussed in Mofidi and Carrica (2014).

Free-running tests in wave basins, as a traditional experimental method, have been recognised as the most robust way to evaluate the manoeuvring performance of ships in waves. This method allows us to understand ships' manoeuvrability in waves with high reliability. However, it is time-consuming, expensive, and technically demanding as well as requires a large wave basin. As a result, ship manoeuvring experiments in waves have been limitedly conducted at a few research institutes. Here are some representative manoeuvring experiments in waves; the Office of Naval Research Tumblehome (ONRT) ship at the Iowa Institute of Hydraulic Research (IIHR) (Sanada et al., 2013; Elshiekh, 2014; Sanada et al., 2019); the KRISO Very Large Crude-oil Carrier 2 (KVLCC2) at Korea Research Institute of Ships and Ocean Engineering (KRISO) (Kim et al., 2019); the KRISO Container Ship at Hiroshima University (Yasukawa et al., 2021).

In the meantime, CFD has been recognised as a high-fidelity numerical method to estimate the manoeuvring characteristics of ships along with the remarkable development of computational power over the recent years. Given the fact that CFD is capable of incorporating both viscous and rotational effects in the flow, it can provide manoeuvring results with higher accuracy, compared to theoretical approaches. At an early stage, several researchers attempted to conduct CFD simulations to understand ships' manoeuvrability in calm water. The manoeuvring analyses of the KCS (Mofidi and Carrica, 2014; Shen et al., 2015), a naval supply vessel (Broglia et al., 2015), and the ONRT ship (Wang et al., 2016) for the calm water condition can be given as good examples. The reliability of the direct CFD approach was demonstrated by the comparative analysis between CFD results and experimental ones. It is worth introducing recent studies where the direct CFD method was widely applied to investigate the manoeuvring performance of the ONRT ship in waves, especially for course keeping manoeuvres (Wang et al., 2017), zigzag manoeuvres (Wang et al., 2018), and turning circle manoeuvres (Wang and Wan, 2018). In addition, Kim et al. (2021b) performed free-running CFD simulations to study the manoeuvring behaviour of the KCS in waves of different propagation directions, focusing on the course-keeping and turning capabilities. Kim et al. (2021a) further extended the application of the free-running CFD model to investigate the manoeuvring behaviour of the KCS in waves of different wave lengths. However, the literature reported here focused on case-specific analyses meaning that research findings from those studies had some inherent limitations in terms of providing a general understanding of the relation between wave heights and a variety of ship types.

In recent years, an increasing demand for understanding a ship's manoeuvrability in waves has drawn more and more attention from academic and industrial audiences (IMO, 2014; ITTC, 2017). However, to date, current research on the effect of wave heights on a ship's manoeuvring performance has been highly limited in number as well as in scope. Extensive experimental and numerical investigations on the manoeuvrability of a vessel in waves were performed within the European funded Project SHOPERA, as presented in Papanikolaou et al. (2015, 2016), el Moctar et al. (2016) and Sprenger et al. (2016). In their research, the manoeuvring characteristics of the KVLCC2 and DTC ship models were comprehensively investigated in different wave conditions (parameters of variation: wave direction, wave period, wave height), contributing to the establishment of a benchmark and validation database regarding manoeuvring problems in waves. Kim et al. (2019) carried out a series of free-running experiments to investigate the manoeuvring characteristics of the KRISO Very Large Crude-oil Carrier 2 (KVLCC2) under three different wave height conditions. The key

Table 1
Principal particulars of the KCS.

Main particulars	Symbols	Model scale (1:75.24)
Length between the perpendiculars	L_{BP} (m)	3.057
Length of waterline	L_{WL} (m)	3.0901
Beam at waterline	$B_{W.L}$ (m)	0.4280
Draft	D (m)	0.1435
Displacement	Δ (m^3)	0.1222
Block coefficient	C_B	0.651
Ship wetted area with rudder	S (m^2)	1.6834
Longitudinal centre of buoyancy	% $L_{BP,}$ fwd+	-1.48
The metacentric height	GM (m)	0.008
Radius of gyration	K_{xx}/B	0.49
Radius of gyration	K_{yy}/L_{BP}, K_{zz}/L_{BP}	0.25
Propeller diameter	D_P (m)	0.105
Propeller rotation direction (view from stern)		Clockwise
Rudder turn rate	(deg./s)	20.1

findings of the previous studies were that the ship's manoeuvring behaviours are greatly influenced by the impact of wave heights. In this regard, this study raised a further question on the relation between wave heights and various types of ships. It is because the manoeuvring characteristics of a ship obviously can vary depending on the types and dimensions of subject vessels. Therefore, in order to draw more generalised and meaningful information, it was highly thought that the manoeuvring characteristics of other ship types under different wave height conditions should be examined. Given the lack of previous research on wave effects on a ship's manoeuvrability, the research reported in this paper was motivated to investigate the effect of wave heights on the manoeuvring performance of the KRISO Container Ship (KCS) model; which has been used in a wide range of research studies but not studied for the manoeuvring behaviours in different wave height conditions. Therefore, this research offers better insight into a container ship's manoeuvrability in a real seaway. It is expected the results generated will be appliable to most conventional container ships.

2. Methodology

This section will provide details of the research methodology used in this study. This research adopted the methodology and the Unsteady Reynolds-Averaged Navier-Stokes model developed by Kim et al. (2021a), who conducted the manoeuvring analysis of the KCS in different wave length conditions. This free-running CFD model has the benefit of being easily extended to various environmental conditions for new applications. As depicted in Fig. 1, the research procedure consisted of four key steps: *(1) goal and scope, (2) numerical modelling, (3) execution of free running simulations, and (4) results*. The first step was designed to set the research objective and define the scope of the study. The second step deals with the numerical modelling of the direct CFD model for the free running manoeuvres in various wave conditions. In the third step, the ship's manoeuvring behaviours are estimated by the free running simulations: self-propulsion, course keeping control, and turning circle manoeuvres. In the fourth step, the general relationship between the ship's manoeuvrability and wave heights are presented through graphs and tables as key findings. The applied methodology will be briefly described in Sections 2.1– 2.4, and the details can be found in Kim et al. (2021a).

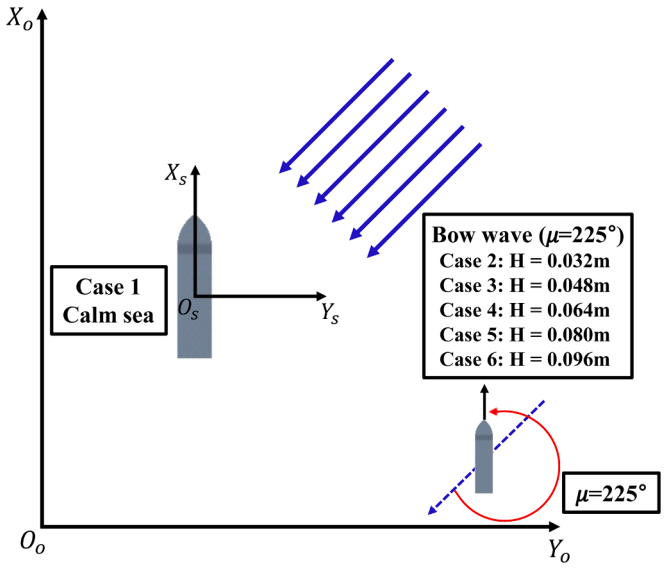


Fig. 2. A schematic view of the simulation cases.

2.1. Step 1: goal and scope

This paper aims to investigate the effects of wave heights on a ship's manoeuvrability in waves. Although several manoeuvring tests are identified to assess the ship's manoeuvring characteristics by ITTC (2021), this paper focuses on the self-propulsion, course-keeping control, and turning capability in various wave heights. The ship type and wave environmental conditions used in the numerical analyses are presented as below.

Throughout all the simulations, the KRISO Container Ship (KCS) model has been used for the manoeuvring analysis in waves. The principal particulars of the KCS are given in Table 1. As seen from Table 2 and Fig. 2, this study designed six manoeuvring simulations: one for the calm water and the others for the waves. As for the wave conditions, the effects of incident regular waves with the wave heights $H = 0.032$ m, 0.048 m, 0.064 m, 0.080 m, and 0.096 m were investigated during free-running manoeuvres (the wave length to ship length ratio $\lambda/L_{BP} = 1.0$ and wave encounter angle $\mu = 225°$). The wave heights considered are ranged from 2.4 m to 7.2 m in full-scale. For the approach speed, it was

Table 2
The simulation cases to which the CFD model is applied.

Case no. C	Approach speed U_0 (m/s)	Propeller rev. (RPS)	Wave height H (m)	Encounter Angle μ (degrees)	Encounter Period T_e (s)	Wave steepness H/ λ	Wave/ship length λ/L_{BP}
1	1.094	13.38	Calm water	- (Calm sea)	–	–	–
2	1.004	13.38	0.032	225 (Bow quartering sea)	1.055	0.010	1.00
3	0.945	13.38	0.048	225 (Bow quartering sea)	1.071	0.016	1.00
4	0.876	13.38	0.064	225 (Bow quartering sea)	1.087	0.021	1.00
5	0.811	13.38	0.080	225 (Bow quartering sea)	1.104	0.026	1.00
6	0.737	13.38	0.096	225 (Bow quartering sea)	1.123	0.031	1.00

found to vary depending on the wave height under the same propeller speed (13.38 RPS), which resulted from the differences in the added wave resistance acting on the ship. Since Kim et al. (2021b) revealed that course-keeping capabilities could be clearly evaluated in oblique seas, this paper has chosen bow quartering waves (μ = 225°). It should be noted that the conducted free-running simulations were subjected to deep water conditions.

2.2. Step 2: numerical modelling

2.2.1. Governing equations

An Unsteady Reynolds-Averaged Navier-Stokes (URANS) method was used to solve the governing equations such as these mass and momentum conservation by employing the commercial CFD software STAR-CCM+, version 15.04. The averaged continuity and momentum equations for unsteady incompressible flows are expressed in tensor notation and Cartesian coordinates as follows (Ferziger and Peric, 2020):

$$\frac{\partial\left(\rho\overline{u}_i\right)}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial\left(\rho\overline{u}_i\right)}{\partial t} + \frac{\partial}{\partial x_i}\left(\rho\overline{u}_i\overline{u}_j + \rho\overline{u'_i u'_j}\right) = -\frac{\partial\overline{p}}{\partial x_i} + \frac{\partial\overline{\tau}_{ij}}{\partial x_i} \tag{2}$$

where ρ indicates the fluid density, $\overline{u}_i$ indicates the averaged velocity vector, x_i (i = 1, 2, 3) are the Cartesian coordinates, $\rho\overline{u'_i u'_j}$ is the Reynolds stresses, $\overline{p}$ is the mean pressure and $\overline{\tau}_{ij}$ are the mean viscous stress tensor components. This stress tensor for a Newtonian fluid can be shown in Eq. (3)

$$\overline{\tau}_{ij} = \mu\left(\frac{\partial\overline{u}_i}{\partial x_j} + \frac{\partial\overline{u}_j}{\partial x_i}\right) \tag{3}$$

in which μ means the dynamic viscosity. The RANS method for turbulent flows requires that the Reynolds stresses in Eq. (4) be appropriately modelled. A common method uses the Boussinesq hypothesis.

$$-\rho\overline{u}_i\overline{u}_j = u_t\left(\frac{\partial\overline{u}_i}{\partial x_j} + \frac{\partial\overline{u}_j}{\partial x_i}\right) - \frac{2}{3}\left(\rho k + u_t\frac{\partial\overline{u}_k}{\partial x_k}\right)\delta_{ij} \tag{4}$$

where the u_t is the turbulent eddy viscosity, and k and δ_{ij} are the turbulence kinetic energy and Kronecker delta, respectively.

In order to model the flow of an incompressible viscous fluid, the RANS solver uses a finite volume method which discretises the integral formulation of the governing equations. The continuity and momentum equations were solved in a segregated manner with a predictor-corrector approach. A second-order upwind scheme was applied for convection and diffusion terms, while a second order temporal discretisation was used for the time domain solution. A semi-implicit method for pressure-linked equations (SIMPLE) algorithm was used to resolve the pressure-velocity coupling.

2.2.2. Physics modelling

To complete the RANS equations, this paper adopted the Shear Stress Transport (SST) k−ω turbulence model (Menter, 1994). The Menter's SST model blends the k−ω and k-ε turbulence models; therefore, the k−ω turbulence model can be used in the inner region of the boundary, and the k-ε model turbulence model can be used in the far field. This turbulence model has been widely used in several past studies in the same area (Mofidi and Carrica, 2014; Wang et al., 2018; Liu et al., 2020; Kim et al., 2021a, 2021b). In order to simulate the motion of the ship in response to pressure and shear forces in the computational domain, the

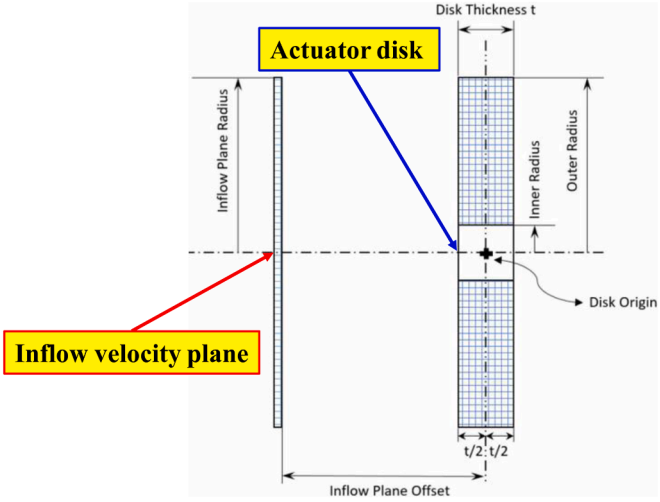


Fig. 3. The schematic view of the actuator disk model and inflow velocity plane, adapted from Siemens (2020).

Dynamic Fluid Body Interaction (DFBI) scheme was applied with the vessel free to move in 6° of freedom (6 DOF). The DFBI model enables the RANS solver to calculate the hydrodynamic forces and moments acting on the rigid body, and to solve the governing equations of rigid body motion to determine the new position of the rigid body (Siemens, 2020). The "Volume of Fluid" (VOF) technique coupled with the 6 DOF solver was applied to model and position the free surface (air-water interface). This VOF method can capture the free surface flow around a ship in either a flat or regular wave.

The actuator disk was modelled to resolve the flow field interaction between the ship hull and the propeller based on the body force propeller method. The propeller experimental results conducted by Hiroshima University (SIMMAN, 2020) were fed into the disk model. In order to model the propeller performance realistically, the propeller-induced velocity was adjusted based on the induced velocity correction property proposed by Neitzel et al. (2015). As shown in Fig. 3, the inflow velocity plane should be designed to calculate the volume-averaged inflow velocity which is required to compute the advance ratio (J). The inflow velocity plane is located upstream of the actuator disk. The advance ratio (J) based on the body force method is expressed as follows:

$$J = \frac{V_{Inflow\ velocity\ plane}}{nD} \tag{5}$$

in which $V_{Inflow\ velocity\ plane}$ represents the volume-averaged velocity over the inflow velocity plane, n is the rotation rate, D is the disk diameter. The advance ratio is then used to determine the operating point from the propeller characteristics curve yielding the thrust and torque coefficients. It is worth noting that Dubbioso et al. (2016, 2017) stated that an actuator disk model needs to be capable of correctly predicting the side forces originated by the propeller to have a good estimation of ship dynamic response. The present CFD simulation adopted the fifth-order Stokes wave model to generate regular nonlinear waves; However, it is thought that the regular waves generated in this work can also be well treated by the first-order wave model as the waves are not steep enough.

2.2.3. Coordinate systems

As described in Fig. 4, four different coordinate systems were introduced in CFD to estimate the ship manoeuvring performance in waves: (1) Earth-fixed frame ($o_o - x_o y_o z_o$); (2) Ship-fixed frame ($o_s - x_s y_s z_s$); 3) Propeller-fixed frame ($o_p - x_p y_p z_p$); and (4) Rudder-fixed frame ($o_r - x_r y_r z_r$). The fluid flow equations were solved, and the hydrodynamic forces and moments acting on the ship were calculated in the earth-fixed inertial reference system. Then, the forces and moments were converted

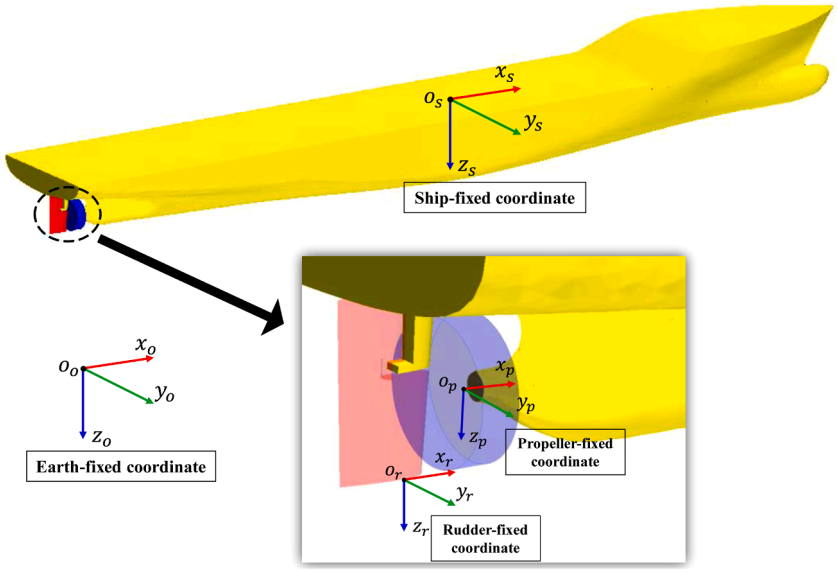
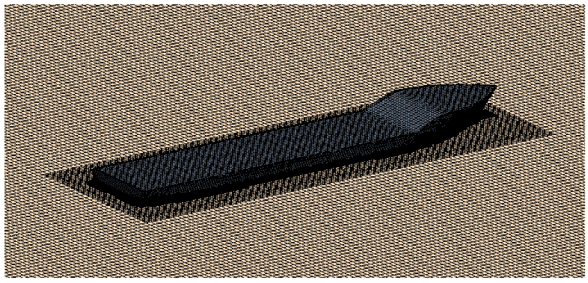
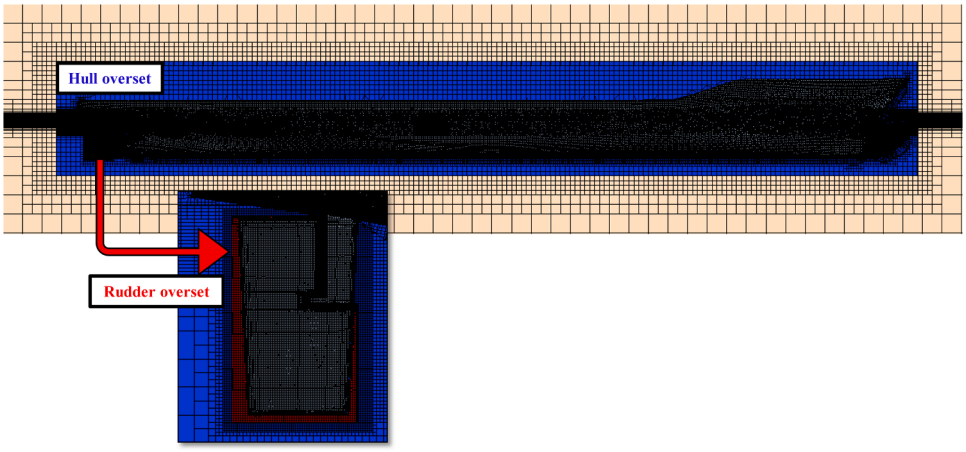


Fig. 4. The coordinate systems of the ship manoeuvring simulation used in this study.



a) Mesh for the ship and free surface



b) Profile view of the computational domain

Fig. 5. Mesh structure of the computational domain.

to the moving ship-fixed reference system. The rigid-body motion equations were solved in the moving frame and consequently the vessel's velocities and new location were calculated. The propeller-fixed frame was defined to use the body force method, while the rudder-fixed frame was introduced to control the rudder movement according to a feedback controller.

2.2.4. Mesh generation

The process of mesh generation was conducted by the built-in meshing facility in STAR-CCM+ to generate hexahedral dominant meshes. A trimmed cell mesher was used to create the volume grids while a surface remesher was applied to obtain a high-quality surface mesh. The ship hull surfaces were covered with six layers of prismatic cells to resolve the surface boundary layer by applying a prism layer mesh model. In order to ensure that the $y+$ value on the ship hull was maintained at a value greater than 30, the height of the first layer mesh was set to be 2.4×10^{-5} m. Fig. 5 presents the final computational domain mesh for the ship manoeuvring simulations in waves. Refined meshes were applied to the vicinity of the ship hull, the tight gaps between rudder blade and horn, and the free surface region to improve the

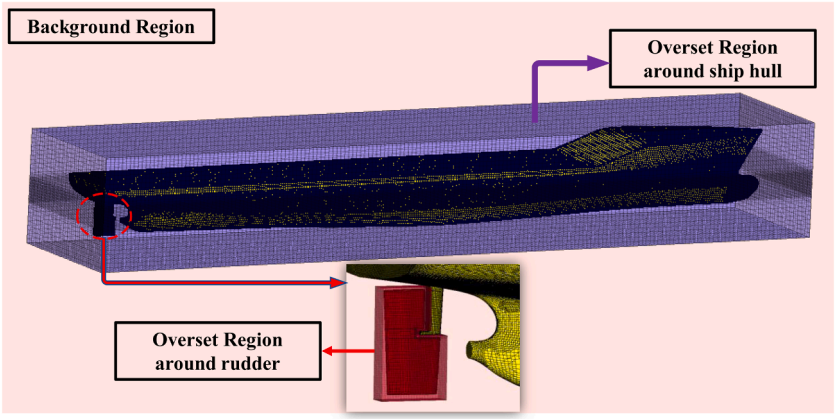


Fig. 6. The schematic of computational domain.

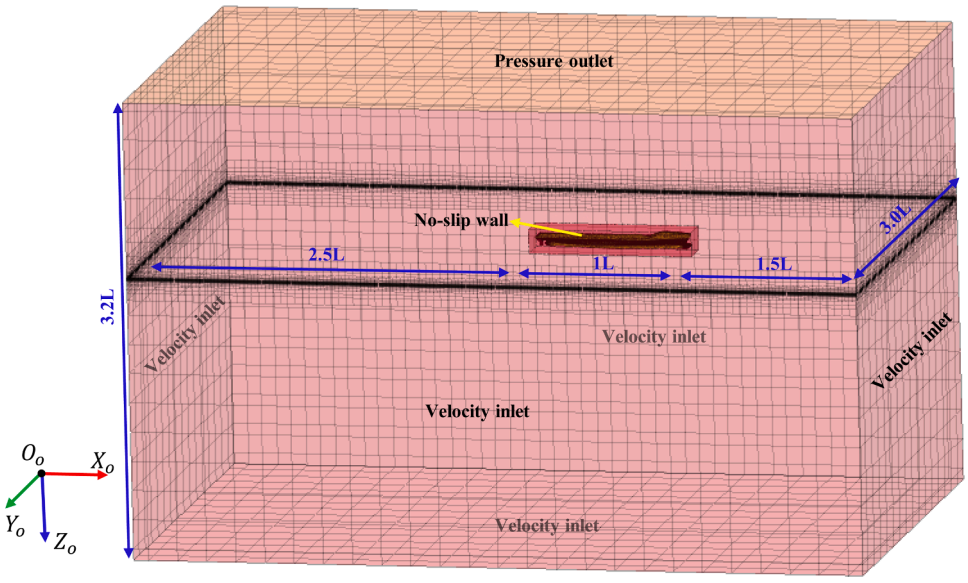


Fig. 7. An overview of the domain with the selected boundary conditions.

accuracy of capturing the flow properties.

The schematic of computational domain is described in Fig. 6. There are three regions in this domain: (1) background region, (2) overset region around the ship hull, and (3) overset region surrounding the rudder blade. The dynamic overset technique was used to carry out free running simulations of the course keeping and turning circle manoeuvres. This overlapping method is capable of handling the complex ship motions and the rudder movement based on the feedback controller. The hole-cutting method in STAR-CCM+ was employed to couple the overset regions with the background region through the overlapping interfaces. The current CFD model was encountered with a challenge to generate the overset grids around the rudder. Since the spacing between the rudder blade (moving part) and the rudder horn is very thin, the overlapping meshes in this area should be fine enough to obtain valid interpolations between the grids.

To simulate ship manoeuvres in waves, the free surface grid refinement was performed based on the practical guidelines for ship CFD applications from ITTC (2011). The recommendation suggests that a minimum of 80 grid points per wave length and 20 grid points per wave height should be generated on the free surface to avoid wave dissipation. In this study's CFD work to ensure a reliable wave propagation during the ship's manoeuvres, a minimum of 80 cells per wave length was used in the x and y directions (horizontal directions). Additionally, a minimum of 20 cells was generated in the z direction (vertical direction) where the free surface was expected. For the calm water simulation, the refined grid area for the free surface was kept relatively small, compared to that used in the wave cases. The total grid number applied to the calm water (case 1) and wave (cases 2–6) simulations is approximately 5.30 $\times$ 10^6 and 8.31 $\times$ 10^6, respectively.

2.2.5. Determination of the time step

For implicit unsteady simulations, it is necessary to determine appropriate time steps to accurately resolve turbulent flow features. Throughout all the simulations, the time step was determined at 0.005 s which is small enough to ensure an accurate description of the wave propagation. This value meets the guidelines of ITTC (2011) that recommend a minimum of 100 time steps per encounter period for wave simulation.

2.2.6. Computational domain and boundary conditions

The boundary conditions of the simulations were defined to represent the free running manoeuvres. Fig. 7 depicts an overview of the domain with the KCS model and the selected boundary conditions. A velocity inlet boundary condition was set in the upstream and downstream boundaries (the positive and negative x directions). The bottom and side boundaries also used the velocity inlet condition to avoid a

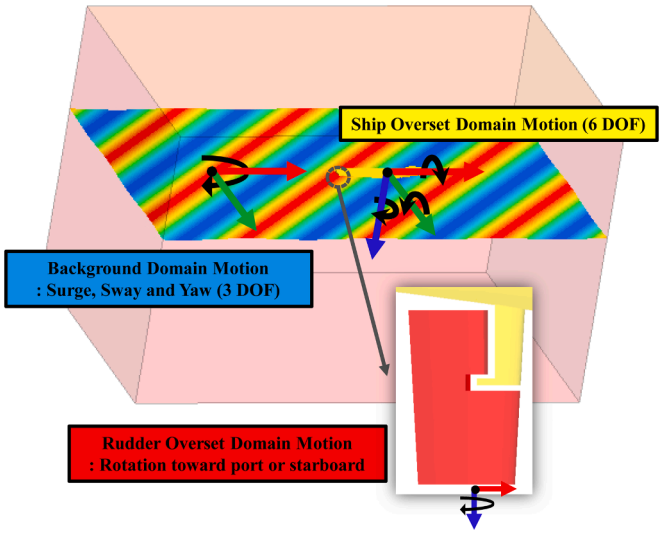


Fig. 8. The view of the motions of the created domains.

velocity gradient from occurring between fluid and the wall. It should be noted that the initial flow velocity at all inlet boundary conditions was set to the corresponding velocity of the incident waves. A pressure outlet condition was imposed at the top boundary at which atmospheric pressure was assumed. For the ship hull and the rudder, no-slip wall boundary conditions were used.

For the manoeuvring simulation in the calm water, the boundaries at the upstream, downstream, and sides used the VOF wave damping capability of the software package with a damping length equal to 1.0 L_{BP} (~3.06 m.). This numerical beach model applied to the vertical boundaries can minimise wave reflections from the walls. For the manoeuvring simulation in the waves, on the other hand, the boundaries at the vertical directions used the wave forcing capability with a forcing length equal to 1.0 L_{BP} (~3.06 m.) from the boundaries. Within the forcing zone, the solution of the discretised Navier-Stokes equations is forced into the theoretical fifth-order Stokes wave solution (Fenton, 1985). This forcing method is also capable of preventing wave reflections from the boundaries. It should be noted that the VOF wave damping and forcing length were selected as recommended by Siemens (2020).

Unlike pure resistance or seakeeping simulations, different motions should be defined for each computational region to simulate free-running manoeuvres. The ship overset region, which contains all grids defined in this region, was solved as the parent level of the six degrees of freedom motion based on the DFBI scheme. The rudder blade and the refinements in the rudder overset region were the children to the ship hull and forced to move according to the rudder controller. The background region, which connects the ship to the far-field boundary conditions, was compelled to follow the ship motion only with respect to the three degrees of freedom (horizontal plane motions: surge, sway, and yaw). Since the background grids are not allowed to pitch, heave and roll, the desired wave propagation can be guaranteed during the simulations. The applied motions for each region are described in Fig. 8.

2.3. Step 3: free running simulations

In order to investigate the effects of the wave height on the ship manoeuvring characteristics, three representative free running simulations were conducted using the direct CFD model: self-propulsion, course keeping control, and standard turning circle manoeuvre.

Achieving the self-propulsion condition at the approach speed should be preceded prior to the course keeping and turning circle manoeuvres. The self-propulsion computations were carried out with six degrees of freedom, with the rudder controlled by a Proportional-Integral-Derivative (PID) feedback controller in order to control the ship's heading angle. Throughout all the simulations, the propeller rotational speed (RPS) was set to be 13.38 n/s. This work applied the same propeller speed as that used for the manoeuvring analysis of the KCS in waves conducted by Kim et al. (2021b). It should be noted that the propeller RPS was maintained constant until completing all free running simulations.

The objective of the course keeping control is to stabilise the yaw angle to a desired heading angle under the wave conditions. Predicting a ship's steering ability is of great importance, since it is closely related to the economy and the safety of ship operations in a real seaway. In order to evaluate the ship's course keeping ability, in this paper, the auto pilot system was modelled using the PID controller. The deflection of the rudder was controlled by the feedback PID controller to keep the ship straight in wave conditions. The controller function is presented as the following expressions:

$$\delta(t) = K_p e(t) + K_i \int_0^t e(t)dt + K_d \frac{de(t)}{dt} \tag{6}$$

$$e(t) = \psi(t) - \psi_c \tag{7}$$

where $\delta(t)$ is the rudder angle, $\psi(t)$ the instantaneous yaw angle at a given time, ψ_c the target yaw angle which was set to be 0° to make the ship advance straight. K_p, K_i, and K_d indicate the proportional, integral, and derivative control gains, respectively. Finding the optimal gains is quite demanding for the good course keeping capability. In this work, the gains were selected to $K_p = 5$, $K_i = 0.05$, *and* $K_d = 3$ after several trials and errors. It is noted that feedback control gains of the autopilot can be changed depending on environmental conditions.

To assess the turning ability of the ship, the standard turning circle manoeuvres were carried out based on the following module:

$$\delta(t) = \begin{cases} \max(0,\ kt), & \delta \le 35 \\ 35 \end{cases} \tag{8}$$

where k is the maximum rudder rate. The maximum rudder rate is set to $k = 20.1°/s$ corresponding to 2.32°/s on full scale. The saturation rudder angle is 35°. From the definitions above, the standard turning manoeuvre turns the rudder to 35° which is maintained constant until the end of the turning simulation.

2.4. Step 4: results of analysis

Step 4 was to estimate the effects of the wave height on the ship manoeuvring performance, namely the self-propulsion, course keeping, and standard turning manoeuvres. By conducting the free running simulations with an extensive range of wave heights, the impacts of the wave height on the ship's manoeuvrability were assessed.

3. Case studies (results of step 4)

3.1. Verification and validation

Similar to most previous studies featuring CFD free running simulations (Mofidi and Carrica, 2014; Wang and Wan, 2018; Wang et al., 2018; Liu et al., 2020), the verification study (spatial and temporal convergence studies) has not been conducted in this paper due to the high computational cost. Mofidi and Carrica (2014) state that the cost of refining and coarsening is highly demanding on overset regions, which may prevent most researchers from attempting a verification study using the direct CFD model. However, verification studies of similar problems using the same CFD model and methodology have been carried out by Kim et al. (2021b) for the manoeuvring behaviour of the KCS in the head waves. According to Kim et al. (2021b), numerical uncertainties for critical manoeuvring quantities are estimated to be a maximum of

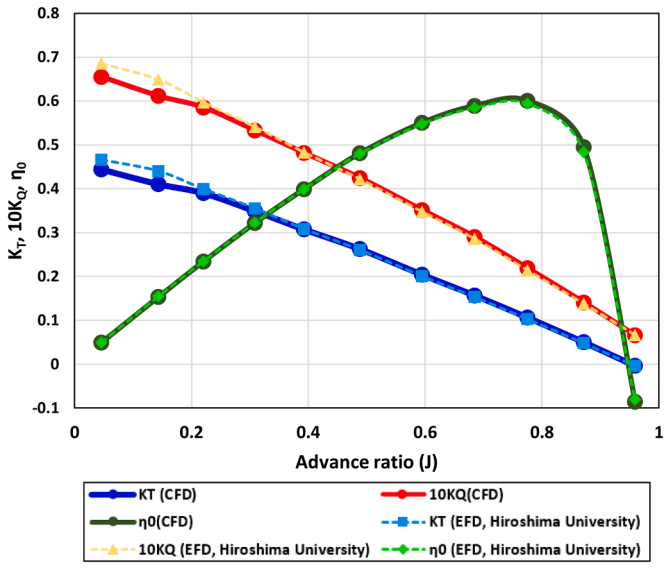


Fig. 9. Propeller open water test results and comparison.

Table 3
Comparison of the main parameters of a standard turning circle manoeuvre in calm water, adapted from Kim et al. (2021b).

Parameters	CFD	EFD Yasukawa et al. (2021)	Error (%)
RPS at self-propulsion point	10.56	10.40	1.58
Advance (m)	9.30	9.29	0.01
Transfer (m)	3.95	4.16	4.86
Time for yaw 90° (s)	15.45	15.64	−1.22
Tactical diameter (m)	9.94	9.66	2.89
Time for yaw 180° (s)	31.50	30.50	3.28

0.28% in the grid convergence test and 0.19% in time-step convergence study, based on Grid Convergence Index (GCI) approach.

The open water calculation of the actuator disk is performed before the free running manoeuvres. During the calculation, the revolution rate of the actuator disk is set to be 12.5 n/s according to the experimental setup from Hiroshima University. The thrust and torque of the actuator disk are computed for a wide range of advance coefficients J from 0.05 to 0.95. Thrust coefficients K_T, torque coefficients K_Q, and efficiency η_0 for each advance ratio J are predicted from the obtained thrust and torque. The CFD calculation results of the open water curves together with the experimental data are presented in Fig. 9. In general, the agreement between the CFD predictions and EFD measurements is very good in terms of thrust coefficient, torque coefficient, and efficiency, with some underprediction of the thrust and torque coefficients for $J = 0.15$. A possible reason for the underestimation by CFD of the K_T and K_Q for $J = 0.15$ is associated with the volume-averaged velocity over the inflow velocity plane. As mentioned in Section 2.2, the thrust and torque acting on the actuator disk are determined based on the computed advance ratio (J). In order for the actuator disk to yield the same K_T and K_Q as the EFD, the disk model should predict the same advance ratio (J) as the EFD under the same experimental condition. Given that the advance ratio (J) of the actual disk is calculated by Eq. (5), it is thought that the volume-averaged velocity over the inflow plane is the main parameter affecting the propeller model performance. It is seen from the CFD results that the actuator disk experiences a relatively larger inflow velocity for the experimental condition ($J = 0.15$) and leads to the lower K_T and K_Q (which implies that the advance ratio acting on the actuator disk model is greater than $J = 0.15$). This may be because the actuator disk model based on the body force method exerts a relatively stronger

Table 4
Comparison of the main parameters of a standard turning circle manoeuvre in head wave, adapted from Kim et al. (2021b).

Parameters	CFD	EFD Yasukawa et al. (2021)	Error (%)
RPS at self-propulsion point	13.38	13.2	1.36
Advance (m)	7.86	8.20	-4.11
Transfer (m)	3.09	3.35	-7.78
Time for yaw 90° (s)	12.52	13.29	-5.81
Tactical diameter (m)	7.97	8.37	-4.83
Time for yaw 180° (s)	23.74	25.32	-6.24

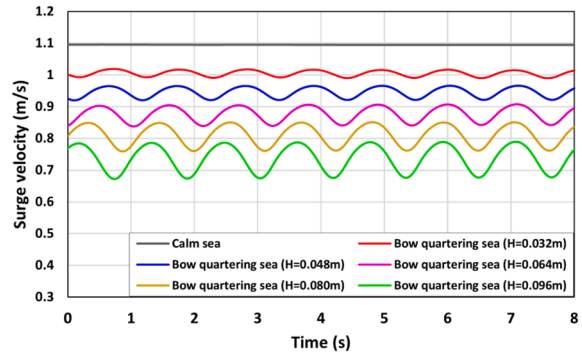
influence on the flow velocity around the inflow plane when the load is high (i.e., $J = 0.15$), consequently resulting in a relatively larger inflow velocity. Further research is needed to better understand the reasons for this discrepancy. Given a good agreement between CFD and EFD results in most cases, it can be claimed that the actuator disk based on the body force method is capable of estimating the propeller performance in terms of thrust and torque. Thus, the virtual disk model is applied for all the free-running manoeuvres.

Unfortunately, experimental results for this ship manoeuvre in the bow quartering waves are not available in the literature, and thus could not be discussed in this paper. However, the current CFD model for free-running simulations has been validated to large extent under various ship hydrodynamic and manoeuvring features in calm water and head waves by Kim et al. (2021b), as partially described in Tables 3 and 4. The agreement between CFD predictions and experimental results available is highly satisfactory with tolerable errors up to 5% in calm sea and 8% in head seas, respectively. More detailed information about the comparisons can be found in the study of Kim et al. (2021b). Given this, the present CFD method can be argued to be validated and can be applied for further investigations. This model is therefore applied throughout all the manoeuvring simulations.

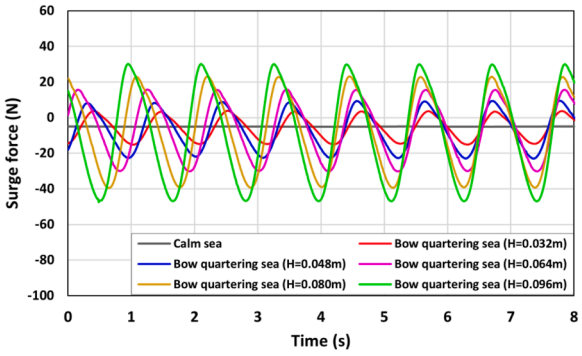
3.2. Self-propulsion

Prior to the investigation of the ship manoeuvrability in waves, the self-propulsion condition should be initially achieved with the ship free to move in 6° of freedom. Fig. 10 presents the time histories of the obtained approach speed, ship resistance, pitch motion, and heave motion at the self-propulsion condition for all the cases. Fourier Series (FS) harmonic analysis is used to analyse the unsteady time histories of the CFD results due to wave motions. The 0th harmonic term in FS means the average value of the time history of the obtained results, whilst the 1st harmonic term refers to the mean amplitude of the oscillation of the values. The FS results for the force and motions are presented in Table 5.

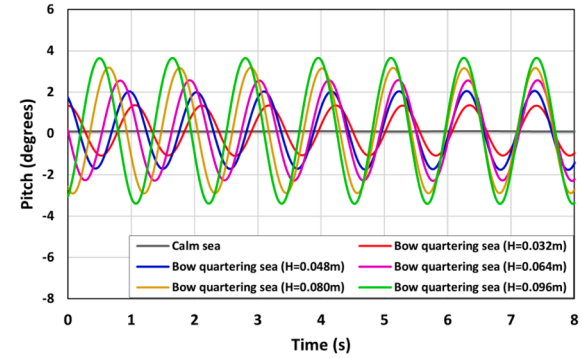
As Fig. 10 and Table 5 jointly show, the ship performance at the self-propulsion conditions shows a significant variation on wave heights. It should be reminded that regular waves with the ratio of wave length to ship length, λ/L_{BP}=1.0, and wave encounter angle, $\mu = 225°$, are applied throughout all the wave simulations, while only changing the wave heights. The wave height ranges from 0.032 m to 0.096 m, corresponding to 2.4 m to 7.2 m in full scale. As seen in Fig. 10(a), the increase of the wave height leads to a decrease in the average approach speed. The mean forward speed in calm water is observed at 1.094 m/s, which corresponds to 18.45 knots in full scale and a Froude number of 0.20. The reduction of the mean approach speed is predicted to be 8.2% for the $H = 0.032$ m, 13.6% for the $H = 0.048$ m, 19.9% for the $H = 0.064$ m, 25.9% for the $H = 0.080$ m, and 32.6% for the $H = 0.096$ m, compared to the calm water case. This is because the ship resistance is found proportional to the wave height, as described in Fig. 10(b). It is worth noting that the ship resistance has a proportional relation with the added wave resistance being commonly proportional to the wave height squared. The oscillation amplitude for the forward speed and ship resistance is significantly increased as the wave heights grow.



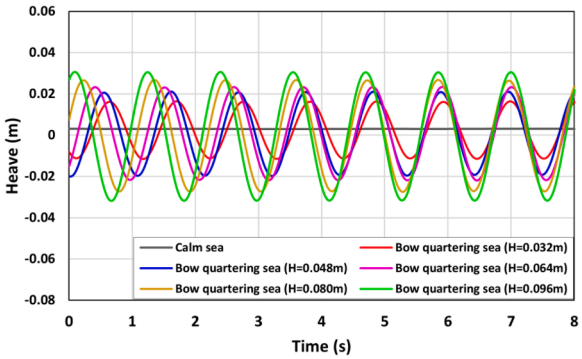
(a) Approach speed (U_0)



(b) Ship resistance (F_x)



(c) Pitch displacement



(d) Heave displacement

Fig. 10. Time histories of the (a) approach speed, (b) ship resistance, (c) pitch motion and (d) heave motion for all the cases.

As for the pitch and heave motions depicted in Fig. 10 (c) and (d), it is revealed that the amplitude of the ship motions in waves varies significantly depending on the encountering wave height. The largest pitch motion is predicted at the bow quartering waves of H=0.096m with a mean amplitude of 3.37°, which is 2.5 times greater than that at the bow quartering waves of H=0.032m. This means the pitch excitation moment increases with the growth of the wave height. The heave motion also follows the same trend. The maximum heave motion is also observed at the bow quartering waves of H=0.096m with a mean value of 0.03m, which is 2 times larger than that at the bow quartering waves of H=0.032m. It should be noted that the ship experiences large amplitude ship motions combined with severe bow slamming under large wave height conditions, which could lead to hull damage. Throughout all the wave simulations, the pitch and heave motions show a similar trend in the ship motion frequency since the same wave length condition (λ/L_{BP} =1.0) is applied to all the wave cases. In addition, the heave responses are non-dimensionalised by wave amplitude (H/2), whereas the pitch responses are non-dimensionalised by wave steepness (H/ λ) in Table 5. As seen in the table, Case 2 and Case 3 appear to have relatively large non-dimensional motion amplitudes, compared to the rest of the cases. This can be explained by the fact that the encounter frequencies of Case 2 (f_e=0.947Hz) and Case 3 (f_e=0.934Hz) are close to the natural frequency of the heaving and pitching system ($f_n \approx$ 0.93Hz), as observed in Kim et al. (2021b).

3.3. Course keeping control

Given that, in general, ships have planned schedules, voyage plans could be determined with a careful route arrangement in consideration of critical safety factors related to route characteristics before a voyage commences. A navigation route in the voyage plan is separated into different straight-line segments, each of which is denoted by the true course between two consecutive waypoints. In navigation practice, the true course can generally be regarded as the target heading angle to be set as input in the auto-pilot system, implying that course keeping control using the auto pilot can be identical to automatic heading control. Hence, a vessel sailing at sea may be deviated from its original course when the heading angle deviation from the target heading angle occurs. In this section, the course keeping capabilities of the ship in waves are evaluated based on this automatic heading control mode (as described in Section 2.3). It is also worth mentioning that recently developed auto pilot systems are capable of correcting the ship's course to return to the original course, using the new route control function in which a route is automatically created between the current position and destination (though this is not studied in the present work).

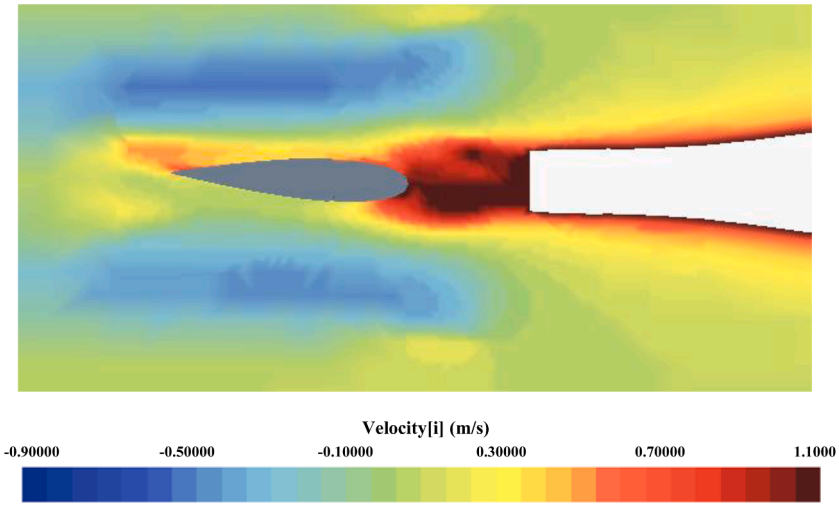
It is true that a vessel operating at sea experiences external forces and moments by environmental disturbances. In particular, waves largely cause deviations as it is difficult to control the yaw angle of a vessel under wave loads. These deviations can consequently result in an increase in the sailing distance with financial repercussions, as well as serious maritime incidents potentially like grounding and collision. Therefore, a proper understanding of the course keeping capability for a subject vessel in a seaway is of great importance. Recently, IMO (2017) has also introduced the guidelines on the steering capability of ships in adverse weather to ensure navigational safety.

The main benefit of the present CFD model is that it is capable of predicting the course keeping ability of a ship in waves. The direct CFD approach has been used for estimating the course keeping capability of the ONRT (Wang et al., 2017) and the KCS (Kim et al., 2021a, 2021b) in waves. However, these studies only deal with the course keeping manoeuvres in different wave direction or wave length conditions. Thus, this section could provide a better understanding of the course keeping control under various wave height conditions.

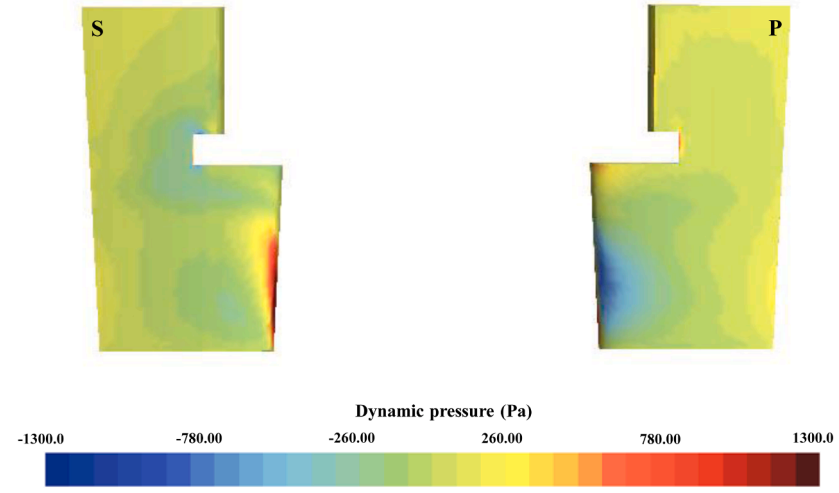
Figs. 12 and 13 present the time histories of the rudder deflection angle, ship's heading angle, and yaw velocity under the course keeping manoeuvres. As seen in Figs. 12(a) and 13(a), it is revealed that the course keeping manoeuvre in calm water is not an issue because there are no external disturbances. The ship's yaw angle and yaw velocity are kept to be almost 0° and 0°/s, respectively, while the rudder deflection is fixed at 2°. This small rudder deflection towards the port side results from a right-handed propeller effect, which causes the asymmetric

Table 5
Fourier series analysis of the approach speed, ship resistance, pitch motion and heave motion at self-propulsion conditions.

	Case 1 (Calm sea)	*Case 2 (H = 0.032 m)*	*Case 3 (H = 0.048 m)*	*Case 4 (H = 0.064 m)*	*Case 5 (H = 0.080 m)*	*Case 6 (H = 0.096 m)*
Approach speed U_0						
0th FS term (m/s)	1.094	1.004	0.945	0.876	0.811	0.737
Resistance F_x						
0th FS term (N)	5.136	5.740	6.723	6.601	7.707	8.931
1st FS term (N)	–	8.583	15.548	21.649	29.102	35.237
Pitch displacement						
0th FS term (degrees)	0.111	0.135	0.170	0.104	0.121	0.203
Non-dimensionalised 0th FS term	–	12.896	10.826	4.967	4.623	6.464
1st FS term (degrees)	–	1.227	1.876	2.402	3.007	3.367
Non-dimensionalised 1st FS term	–	117.212	119.473	114.728	114.900	107.213
Heave displacement						
0th FS term (m)	0.003	0.002	0.002	0.001	0.000	0.000
Non-dimensionalised 0th FS term	–	0.125	0.083	0.031	0.000	0.000
1st FS term (m)	–	0.014	0.019	0.022	0.026	0.030
Non-dimensionalised 1st FS term	–	0.875	0.792	0.687	0.650	0.625



(a) Axial flow velocity around the rudder



(b) Pressure distribution on the rudder blade, S: starboard profile, P: port profile

Fig. 11. The snapshots of the axial flow velocities around the rudder and the pressure distributions on the rudder under the course keeping manoeuvre.

pressure distribution acting on the rudder surface. The resultant uneven pressure distribution induces a small yaw moment to turn the ship's heading to the starboard. Thus, the rudder angle is slightly deflected to the opposite side to control the ship's heading. Fig. 11, as an example, displays the snapshots of the axial flow velocities around the rudder and the pressure distributions on the rudder under the course keeping manoeuvre (Case 1). When the ship is moving forward in calm water, a non-uniform flow generated by the actuator disk causes the pressure

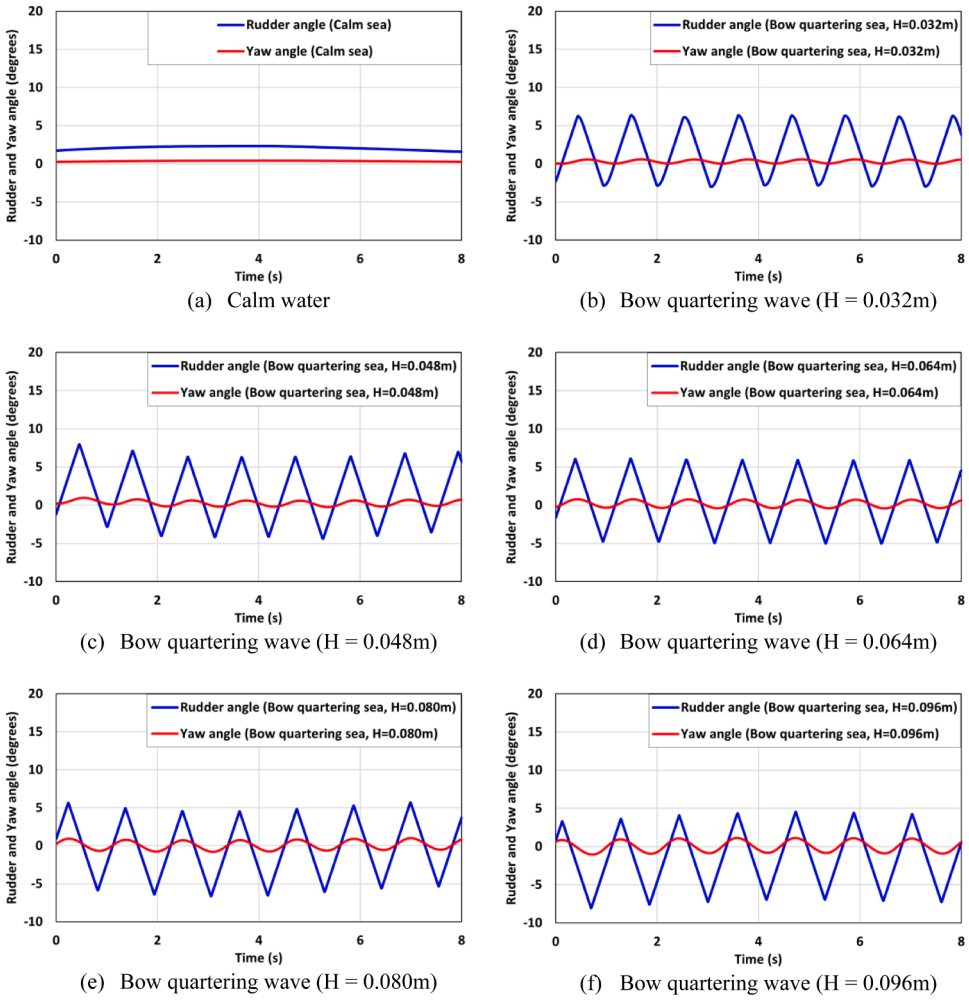


Fig. 12. Time histories of the rudder deflection and yaw angle at self-propulsion in (a) calm sea, (b) bow quartering sea ($H = 0.032$ m), (c) bow quartering sea ($H = 0.048$ m), (d) bow quartering sea ($H = 0.064$ m), (e) bow quartering sea ($H = 0.080$ m), and (f) bow quartering sea ($H = 0.096$ m).

difference between the starboard and the port of the rudder blade. This leads to a small rudder normal force, and thus causes the yaw moment to make the ship turn to the starboard to a small extent. It is worth noting that the actuator disk model accounts for the axial and tangential (swirl) velocities induced by the propeller and its effects on the flow.

It is obvious that the rudder deflection angle becomes larger in the oblique waves compared to the calm water case. This is because an asymmetric wave profile around the ship is generated when the ship is advancing forward in the oblique seas, which may cause a substantial lateral force and yaw moment and consequently induce yaw deviation. During the course keeping manoeuvres in waves, the behaviours of the rudder deflection and yaw angle are found to vary depending on the wave height conditions. Although similar maximum rudder angles are deflected in all bow quartering seas within a value of 7.5° as shown in Fig. 12(b)–(f), the starboard rudder deflection becomes relatively dominant as the wave height increases. It results from the fact that a higher wave height induces a larger yaw moment to change the ship's heading to the port side under the starboard bow quartering waves. Likewise, the oscillation amplitude of the yaw velocities also increases with the growth of wave heights as seen in Fig. 13(b)–(f). The predicted ship trajectories under the course keeping manoeuvres are depicted in Fig. 14. The ship advancing in the calm water achieves good course-keeping control by showing a very small deviation from the original course. For wave cases, all the ship paths are clearly different from the trajectory predicted in the calm sea. The maximum deviation from the original course is noted in the highest wave ($H = 0.096$ m) whereas the minimum deviation is observed in the lowest wave ($H = 0.032$ m). It reveals that higher wave heights can lead to a large deviation from the planned route and thus cause relatively poor performance of the course keeping control. In addition, it is estimated that the high wave height conditions can cause the course of the ship to oscillate significantly during the manoeuvre. It should be noted that the optimum control gains for minimising the yaw deviation can further improve the steering capability.

3.4. *Turning circle manoeuvre*

This section will outline the simulation results of a standard turning manoeuvre. In turning circle manoeuvres, the ship is advancing forward at an approach speed and the rudder is deflected to a maximum 35-degree angle in the starboard direction. It should be noted that the rudder starts to be executed during the moments when the wave trough passes on the midship. The ship starts to react by turning to the starboard side and the manoeuvre is completed when the 360° turn is achieved. This results in turning responses that are used to assess the manoeuvrability of the ship. As seen in Fig. 15, the standard turning circle manoeuvre as recommended by ITTC (2021) measures the manoeuvrability of a ship by obtaining the main indices of the manoeuvre, i.e., the advance, the transfer, the tactical diameter, and the time to 90°/180° yaw angle change. Reference can be made for more detailed information on the turning circle parameters to ITTC (2021).

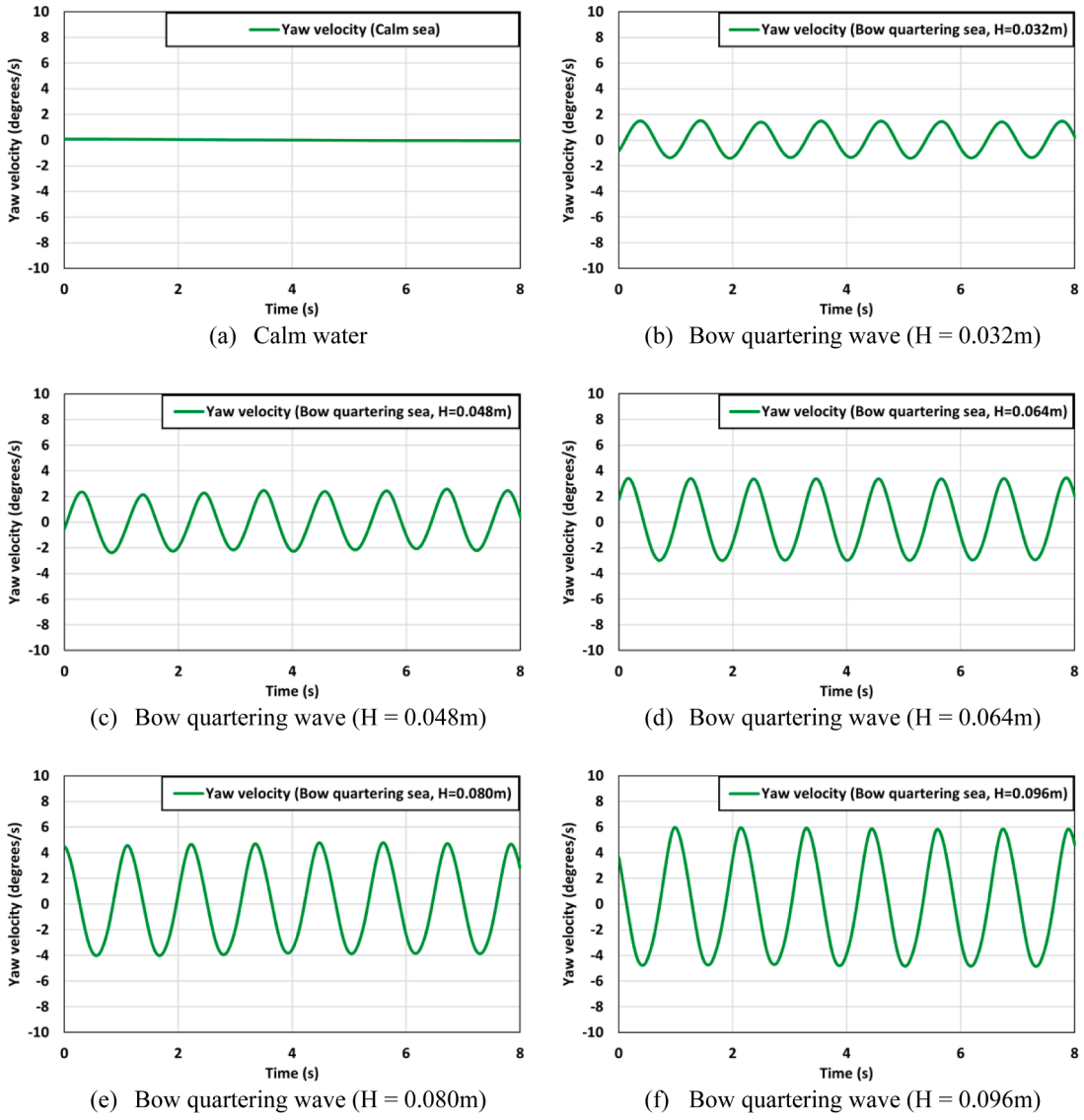


Fig. 13. Time histories of the yaw velocity at self-propulsion in (a) calm sea, (b) bow quartering sea ($H = 0.032$ m), (c) bow quartering sea ($H = 0.048$ m), (d) bow quartering sea ($H = 0.064$ m), (e) bow quartering sea ($H = 0.080$ m), and (f) bow quartering sea ($H = 0.096$ m).

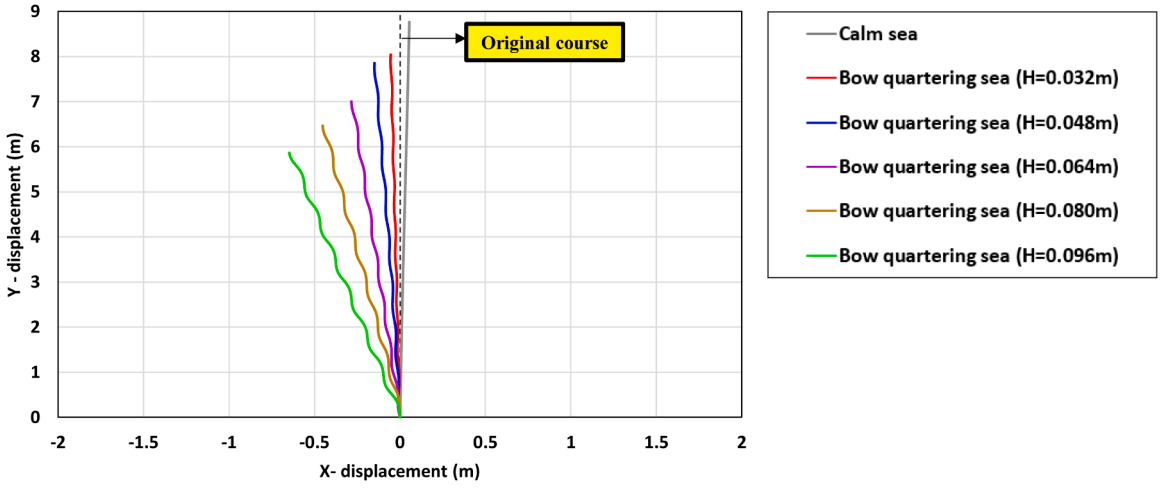


Fig. 14. Comparison of the predicted trajectories for all the cases.

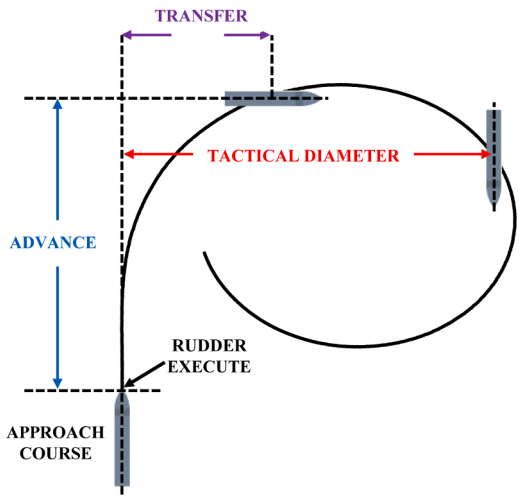


Fig. 15. The definitions of a turning circle.

3.4.1. Time histories during turning and turning indices

In Fig. 16 all the obtained ship trajectories of the turning circle manoeuvre are displayed. In the figure, all the rudder execution points are shifted to the origin point (0,0) for the correct comparisons of the turning trajectories between each case. It is clearly demonstrated that higher wave heights can lead to larger changes in the turning trajectory when compared to the ship's inherent turning ability in calm water. In other words, the significant deformation of the turning circle path occurs under the higher wave height conditions, compared to the calm water case. It is seen that the ship trajectories are drifted in the bow quartering waves as the ship is influenced by continuous wave drift forces during the manoeuvre (the drift directions are observed to be similar to the wave propagation direction).

To quantify the turning capability for all cases, the CFD results in terms of the critical turning indices are reported in Table 6. In addition, Fig. 17 shows the time histories of the predicted ship velocities, forces, and moments during the turning manoeuvre. Table 6 and Fig. 17 jointly show, the maximum ship advance from the origin point is predicted to be $3.13L_{BP}$ in the calm water (case 1) since the approach speed is much larger than the wave cases. For the wave cases (cases 2–6), the advance is found to mainly decrease with the increase in the wave height. This is due to the relatively smaller approach speed despite the longer 90° turning time as the wave height increases. The trend observed for the ship transfer is found similar to the ship advance. The transfer reaches a maximum of $1.33L_{BP}$ under the calm water condition. For the wave conditions, the transfer shows a decreasing trend with the increase of the wave height. This is attributed to the relatively smaller forward and sway velocity in the initial phase of the turn and the longer time taken for 90° turn. The tactical diameters are also clearly identified to be dependent on the wave height. They follow the same trend as the advance and transfer of the case ship, exhibiting a decreasing tendency with the increase of the wave height.

Clearly different from the calm water case, the ship manoeuvring in waves experiences high-frequency fluctuations of the kinematic and dynamic quantities during the turning manoeuvre (Fig. 17), mainly due to the high-frequency wave-induced motions. Larger fluctuation amplitudes occur under the higher wave height conditions, closely associated with the trajectory oscillation experienced by the ship in waves, as seen in Fig. 16.

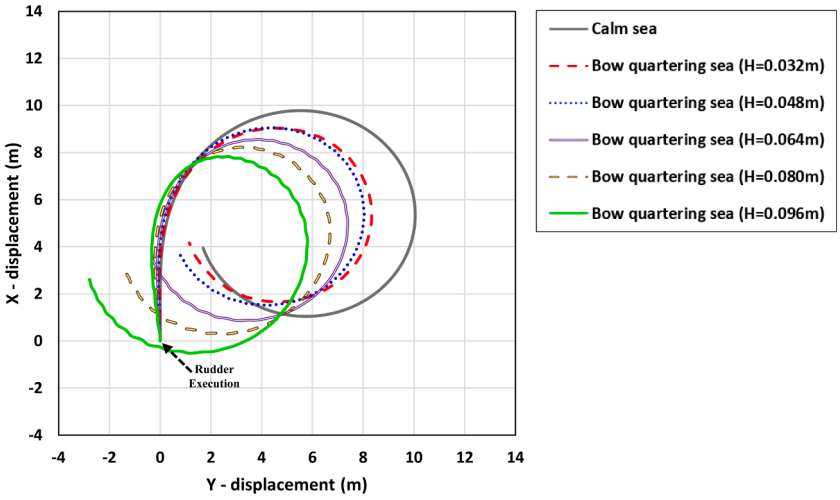


Fig. 16. Comparison of the predicted turning circle trajectories.

Table 6
CFD results: turning indices in calm water and regular waves.

Parameters (CFD results)	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Advance (m)	9.55 **($3.13L_{BP}$)**	8.79 **($2.88L_{BP}$)**	8.83 **($2.89L_{BP}$)**	8.35 **($2.73L_{BP}$)**	8.04 **($2.63L_{BP}$)**	7.70 **($2.52L_{BP}$)**
Transfer (m)	4.07 **($1.33L_{BP}$)**	3.24 **($1.05L_{BP}$)**	3.09 **($1.01L_{BP}$)**	2.68 **($0.88L_{BP}$)**	2.24 **($0.73L_{BP}$)**	1.17 **($0.38L_{BP}$)**
Time for yaw 90° (s)	12.31	12.18	13.08	13.44	14.18	15.10
Tactical diameter (m)	9.82 **($3.21L_{BP}$)**	8.09 **($2.65L_{BP}$)**	7.82 **($2.56L_{BP}$)**	7.17 **($2.35L_{BP}$)**	6.54 **($2.14L_{BP}$)**	5.63 **($1.84L_{BP}$)**
Time for yaw 180° (s)	24.20	23.19	24.38	24.93	26.26	27.31

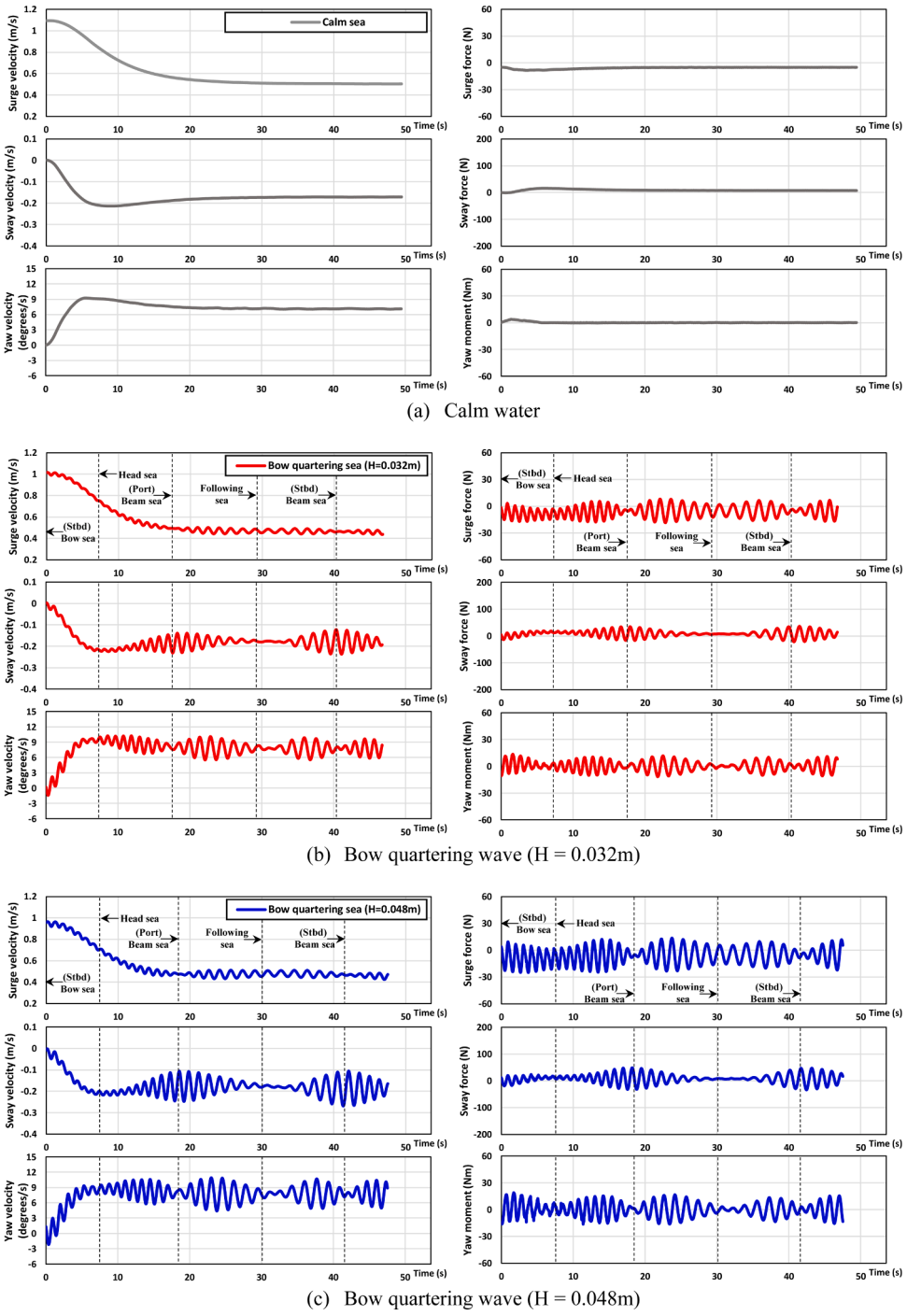
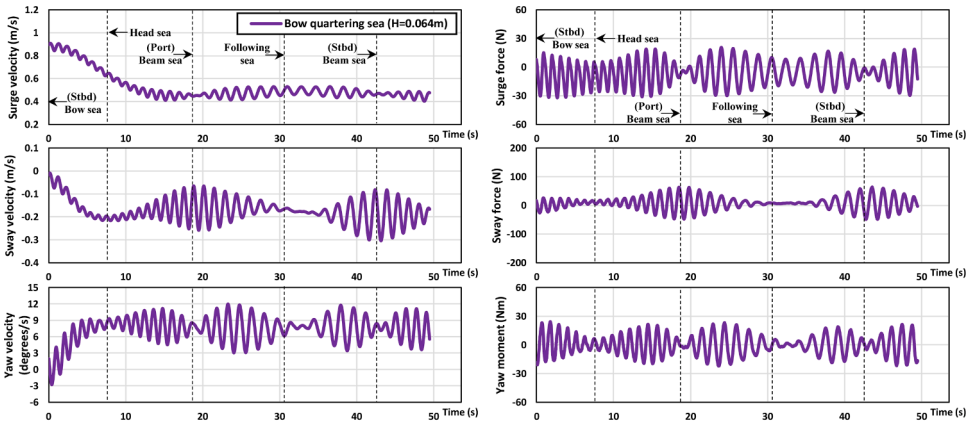


Fig. 17. Time histories of the ship velocities, forces, and moment during a turning manoeuvre in waves.

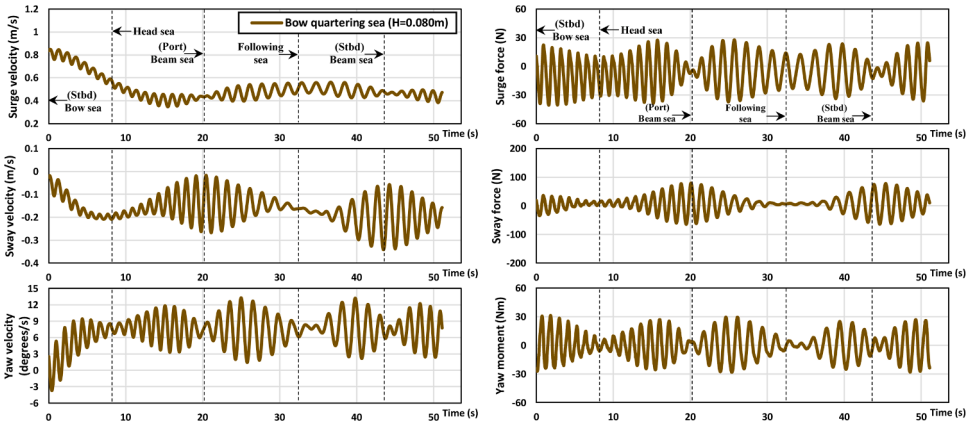
3.4.2. *Wave-induced motions during turning manoeuvre*

This section will deal with the ship motions predicted during the starboard turning circle manoeuvre. Fig. 18 presents the time histories of ship motions, i.e., heave and pitch, as well as pitch moment and heave force. The oscillation amplitudes of the ship motions are found to change continuously during the ship's turning manoeuvre due to the instantaneous variations in the wave-encounter directions. For example, the ship advancing in the bow quartering seas (cases 2–6) experiences the starboard bow quartering wave (0° turn), the head wave (45° turn), the port beam wave (135° turn), the following wave (225° turn), the starboard beam wave (315° turn), and the starboard bow quartering wave (360° turn) in series after the start of the starboard turning manoeuvre. In addition, the wave-encounter frequencies are also observed to continue to change due to the variations in the ship's velocity and the wave-encounter direction during the ship's turning, which consequently affects the ship motion frequencies in waves.

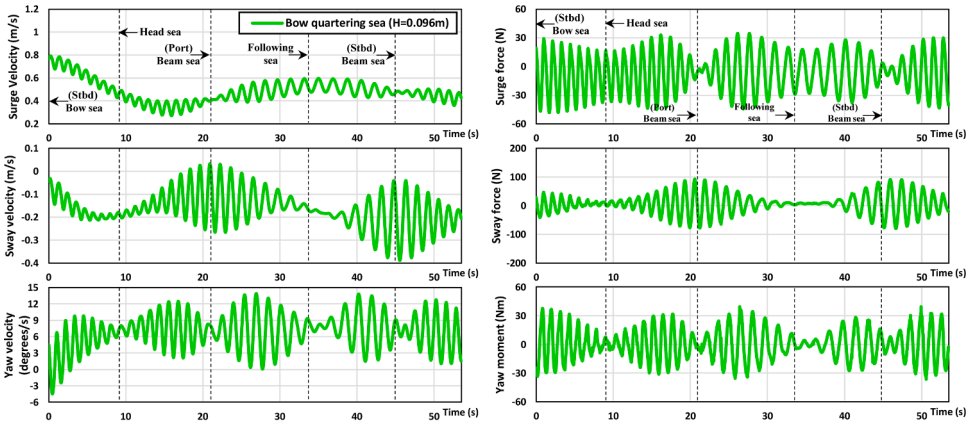
As expected, it is found that the excitation force and moments increase with an increase in the wave height and thus the amplitude of the pitch and heave appears to be larger in the higher wave heights. During the ship's turning, the pitch oscillations reach the maximum when the ship experiences the bow quartering waves whereas the minimum pitch motions are predicted under the starboard or port beam waves. As stated



(d) Bow quartering wave (H = 0.064m)



(e) Bow quartering wave (H = 0.080m)



(f) Bow quartering wave (H = 0.096m)

Fig. 17. (*continued*).

previously, the pitch motions tend to have relatively large amplitudes when the encountering frequency is close to the natural frequency of the pitching system.

Contrary to the pitch motion, the maximum heave amplitudes are observed when encountering the starboard or port beam waves while the ship is turning. This is also closely associated with the ratio of wave length to ship length. As the ship starts tuning in waves, the wave length becomes relatively longer than the ship length. When the ship encounters the beam waves (135° turn or 315° turn), the ship breadth can be regarded as the relevant length which is relatively smaller than the wave length. It may lead to the maximum heave amplitude which is almost equal to the wave height. The minimum heave motions are predicted when the ship encounters the following waves.

4. Conclusions and discussion

Despite increasing demands, past research is still insufficient to provide a full understanding of the ship manoeuvrability in a real seaway. As such, this paper will add a sound value to this field showing the effectiveness of the proposed CFD model to assess the manoeuvrability of the case ship. This study has filled the research gap on the manoeuvring performance of the KCS in different wave heights. The main feature of this research is placed on the extended application of the free running CFD model developed in Kim et al. (2021a), which is

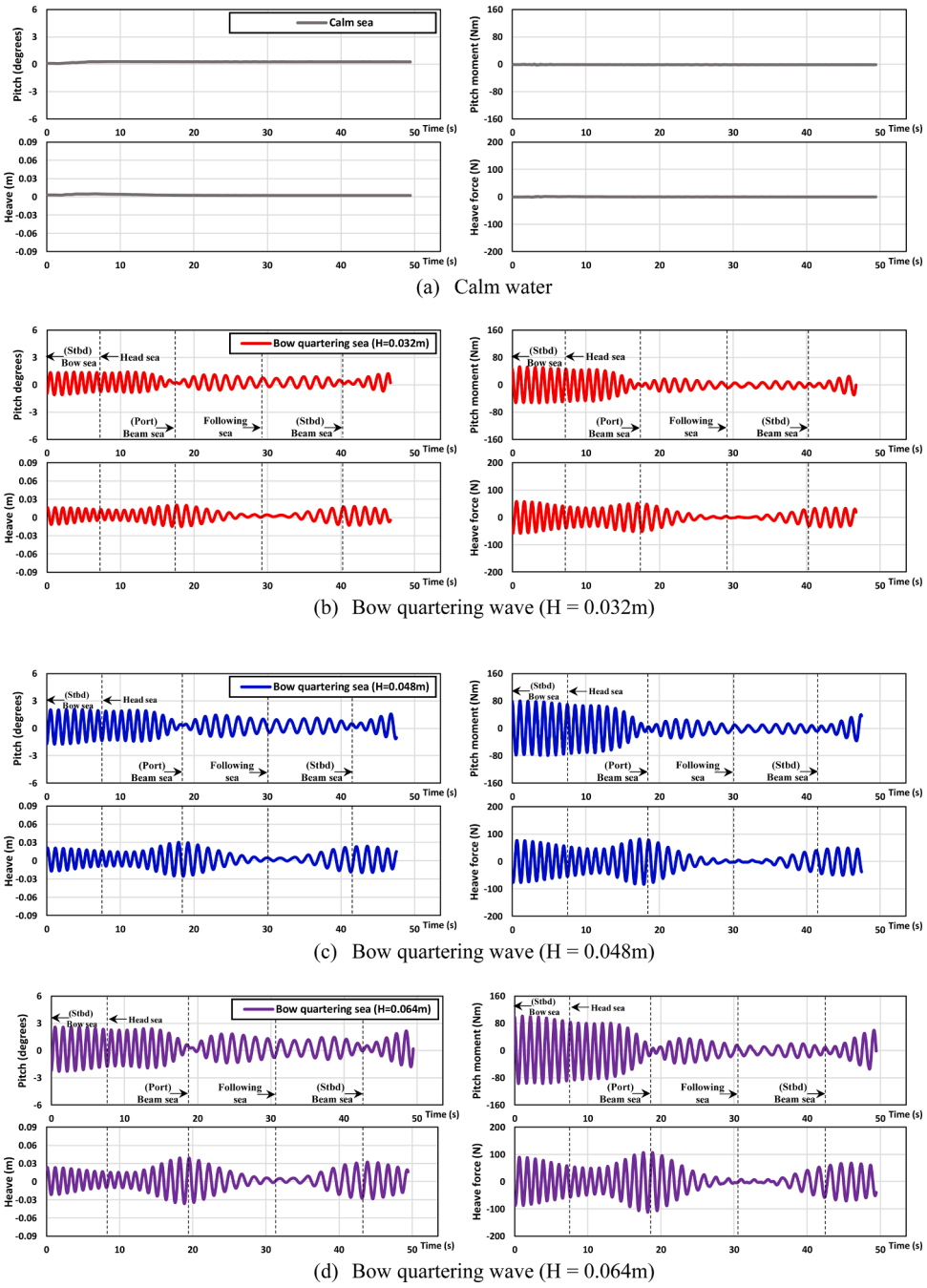
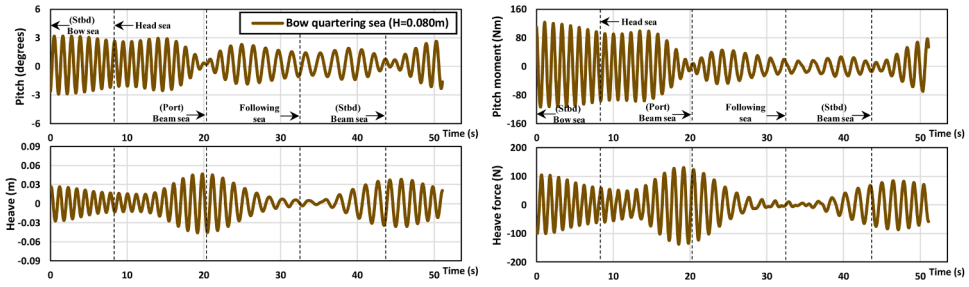


Fig. 18. Time histories of pitch, heave, pitch moment, and heave force during a turning manoeuvre.

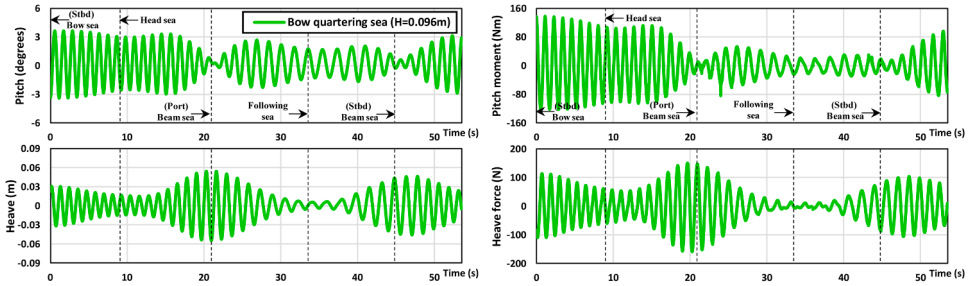
limited to investigating the wave length effects on the ship manoeuvrability. With the analysis of the correlations between the ship manoeuvrability and the wave height, the findings of this study are helpful in identifying the effects of the wave height on the manoeuvring behaviour of the ship in a real sea state. As computational facilities become more powerful and more accessible, this free running CFD method will be an accurate and efficient way to predict the ship's manoeuvrability in waves. The key findings of this study can be summarised as follows:

(1) It is identified that the self-propulsion conditions show significant variation depending on the wave height. The averaged approach speed is found to decrease with the increase of the wave height at a given wave height range ($0.032\ m \leq H \leq 0.096$ m), corresponding to 2.4 m to 7.2 m in full scale. Compared to the calm water case, the maximum reduction of the approach speed is predicted 32.6% for $H = 0.096$ m whereas the minimum reduction is observed to be 8.2% for $H = 0.032$ m under the same propeller revolution. Since the pitch excitation moment increases with the increase of the wave height, the largest pitch motion is predicted for the highest wave. The heave motion also follows the same trend.

(2) For the course keeping manoeuvres, the behaviours of the rudder deflection and yaw angle are strongly affected by the wave height. Although similar maximum rudder angles are observed to be within a value of 7.5° in all bow quartering waves, the starboard rudder deflection becomes relatively dominant as the wave height increases. It is associated with the fact that a higher wave



(e) Bow quartering wave (H = 0.080m)



(f) Bow quartering wave (H = 0.096m)

Fig. 18. (*continued*).

height causes a larger yaw moment to change the ship's yaw angle to the port side under the starboard bow quartering waves. In addition, higher wave heights result in a large deviation from the original course and thus cause poor performance of the course keeping control.

(3) For the standard turning circle manoeuvres, the critical turning indices are found to be directly correlated to the ship velocities in the horizontal plane (surge, sway, and velocities) which differ significantly depending on the wave height. For the wave cases, the ship advance is predicted to mainly decrease with the increase in the wave height. This results from the relatively smaller approach speed despite the longer 90° turning time as the wave height increases. It should be noted that the lower wave height causes a larger yaw moment to turn the ship to the starboard side as the ship starts turning in bow quartering waves, which has an influence on the time to turn by 90° and 180°. The ship transfer and tactical diameter also follow the same trend as the advance. Compared to the calm water case, the turning circle trajectories are identified to be significantly deformed under the higher wave height conditions. The wave forces and moments cause the drift of the ship trajectories, and the drift directions are found to be similar to the wave propagation direction. For the seakeeping behaviour during the ship's turning, the maximum amplitude of the pitch is observed when the ship experiences the bow quartering waves whereas the minimum pitch motions are predicted under the starboard or port beam waves. Contrary to the pitch motion, the heave oscillations reach maximum when encountering the starboard or port beam waves. The minimum heave motions are observed when the ship experiences the following waves.

Considering the ship's manoeuvring performance is greatly affected by the complex interactions between the hull, propeller, and rudder, it is highly believed that propulsion loss may result in considerable changes in the ship manoeuvrability. In particular, the loss of the ship propulsion capability in rough sea conditions could be considered as one of the most serious categories of hazardous events in sea transport. In this context, this research could further be extended by investigating the impacts of the propulsion loss on the ship's manoeuvrability in regular waves as a next step, which will help contribute to enhancing navigation safety at sea. In addition to this, it seems necessary to further perform zigzag manoeuvres in waves, which also has been considered as critical manoeuvring characteristics.

The study should also be extended to investigate the effect of hull roughness on ship manoeuvrability, as the added resistance due to hull roughness will also have a remarkable effect on the manoeuvring behaviour of a ship. The roughness effect can be estimated by using modified wall functions available in CFD software as successfully shown in Song et al. (2020a, 2020b).

In practice, navigational officers have a limited access to the information on the ship manoeuvrability under calm water conditions, which is generally obtained from full-scale sea trials or model-scale experiments. Unfortunately, such data are not equally relevant to the manoeuvrability of the same ship in a real seaway. As the effect of various wave heights on the manoeuvring performance of the KCS has been quantitatively revealed, this paper is believed to have contributed to enhancing navigation safety at sea; which could be a practical insight into the manoeuvring behaviour of commercial vessels in real sea states. The manoeuvring data from this study can assist navigators in determining proper decision-making for ship-handling actions in waves to avoid collision accidents. When vessels in sight of one another involve the risk of collision, navigators should take proper and effective action to avoid collision according to COLREGs (International Regulations for Preventing Collisions at Sea) established by the IMO (2001). COLREGs also states that making a large course alteration may be the most effective action to avoid a close-quarters situation, which obviously leads to the ship's turning behaviour. Thus, it is paramount to fully understand the ship's turning performance when taking action to avoid collision, especially the advance, the transfer, and the time to 90° yaw angle change.

CRediT authorship contribution statement

Daejeong Kim: Conceptualization, Formal analysis, Data curation, Investigation, Writing – original draft, Writing – review & editing. **Soonseok Song:** Formal analysis, Investigation. **Byongug Jeong:** Formal analysis, Investigation. **Tahsin Tezdogan:** Supervision, Writing – review & editing. **Atilla Incecik:** Supervision, Writing – review &

editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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